Hrishikesh Pawar

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Education

Worcester Polytechnic Institute — **August 2023 - May 2025**

Master of Science in Robotics Engineering; **GPA: 3.88/4.0** — *Worcester, MA*

Courses: **CS541**-Deep Learning, **RBE549**-Computer Vision, **RBE595**-Vision based Robot Manipulation, **REB550**-Motion Planning

Teaching Assistant: **RBE595**-Vision based Robot Manipulation, **RBEBL01**-Robots for Recycling

Experience

Pixels on Target | *Computer Vision Engineer* | *Sunrise, FL* — **July 2025 - Present**

- Architected robust perception and sensing algorithms, leveraging **EO/IR** modalities to achieve high fidelity spatial awareness
- Optimized imaging and **sensor fusion** pipelines for real time deployment on resource constrained edge platforms to ensure low latency
- Engineered high precision **spatial synchronization** and **multi sensor alignment** protocols to ensure temporal consistency across platforms

TRUELight Energy | *Machine Learning Intern* | *Boston, MA* — **January 2025 - May 2025**

- Automated workflows across major **ISOs** using **Optical Character Recognition** to extract **energy profiles**, reducing system latency by **50%**
- Deployed **ETL** pipelines on **AWS** (**EC2**) for high throughput data ingestion and energy analytics, cutting operational costs by **40%**
- Formalized system specifications for **TRUEPrice** platform, establishing a standardized framework that accelerated cross-team adoption

Nobi Smart Lights | *AI Software Intern* | *Houston, TX* — **May 2024 - August 2024**

- Led R&D efforts on ceiling-mounted, vision-enabled lamps supporting real-time **fall detection** and emergency response in elderly care
- Engineered a **rotation-aware** detection model by integrating **Swin Transformers,** improving accuracy from **85 mAP** to **92 mAP**
- Explored **LLaVA** and **CLIP** with **LoRA-based PEFT** for fall-context inference, evaluated on **100+** scene-prompt pairs with **61%** baseline score

Adagrad AI | *Computer Vision Engineer* | *Pune, India* — **November 2020 - July 2023**

- Developed core modules for **Gate-Guard**, an **edge-based** system for **vehicle access control** using **Automatic License Plate Recognition**
- Achieved **92%** accuracy for **four-wheelers** and **88%** for **two-wheelers** at **50 FPS** on **NVIDIA Jetson TX2**, with deployment across **100+** sites
- Built real-time dashboards with **Django** and **Azure**, using backend services (**Kafka**, **Celery**, **Redis**, **WebSockets**) to handle **3K+ scans/hour**

Publications

Hrishikesh Pawar, *Deepak Singh, Nitin J Sanket*; Proceedings of the Winter Conference on Applications of Computer Vision (WACV) Workshops, 2025, pp. 912-916 | Publication

- Proposed a bio-inspired navigation strategy that leverages **optical defocus** in **event cameras** for energy-efficient aerial perception
- Modeled passive **sharpness-based** depth cues from defocus patterns to **segment foreground obstacles** from **background regions**
- Achieved **82%** navigation success rate with **62x** computational savings compared to state-of-the-art methods (**Depth-Pro**, **MiDaS**)

Projects

Deep Visual-Inertial Odometry | GitHub | Project Report

- Designed a **VIO** pipeline coupling **CNNs** for visual features and **LSTMs** for **IMU** integration to improve pose estimation accuracy
- Optimized training using **Mean Squared Error** and **Geodesic loss**, resulting in a **28%** reduction in **RMSE** of absolute trajectory error

Zero-Shot Semantic Neural Style Transfer for Images | GitHub

- Implemented AdaAttN module to perform per-point adaptive normalization for **zero-shot semantic neural style transfer** tasks
- Achieved a **13%** training time reduction by removing redundant deep attention layers while maintaining visual style fidelity

Skills

- **Languages**: Python, C, C++, SQL, Bash, MATLAB
- **Frameworks**: PyTorch, OpenCV, TensorFlow, ONNX, NumPy, Blender, SciPy, Scikit-learn, Matplotlib, Pandas, Django, Flask, Celery, Kafka, CMake, Docker, TensorRT, Jenkins, Kubernetes, Git, ROS, ROS2

Leadership and Achievements

- **Presentation**: Presented research on defocus-based depth estimation with event cameras at WACV 2025 Workshop — Link
- **Recipient:** Dr. Glenn Yee Graduate Student Project Award — Link
- **Recognition**: Won **Best Project Award** in **WPI's** RBE-549 Computer Vision course, selected from a cohort of **40** graduate students — Link
- **Leadership**: Led **56-member** team in Formula Student competitions; **rank-1** at Formula Bharat and **top-30** at Formula Student Germany